MATS

The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR A single **linear map** on residual activations predicts an LLM's answer from its context **before generation** held-out R^2 0.75, top-1 retrieval among 1,000 candidates at 0.81 is already ∼87% present in the **base model**, is **shared between the assistant and fictional characters**, and predicts **sycophancy, hallucination, and misalignment before inference**.

MOTIVATION

- LLMs are trained as **next-token predictors**, but the weights fix a deterministic mapping from a context to a distribution over *whole answers*: they are also **next-answer predictors**. Absent tool results, **everything about the answer distribution is fixed before the first token is generated.**
- A **simple approximation** of that mapping would be extremely useful: predict behavior *before* generation, and predict how fine-tuning will change it.
- But it need not exist: the mapping is billions of parameters applying dozens of nonlinear layers, so a priori it could be arbitrarily complex. Prior work sits at two ends of a **capacity–usefulness tradeoff** cheap but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a whole-distribution imitator with no capacity advantage over the model itself.
- **Our question:** can a **very low-capacity** approximation capture the mapping *as a whole*? Such an approximation is a **metamodel**: a much smaller model that reads the LLM's internals and predicts its behavior.

THE RESIDUAL CONTEXT-ANSWER MAPPING

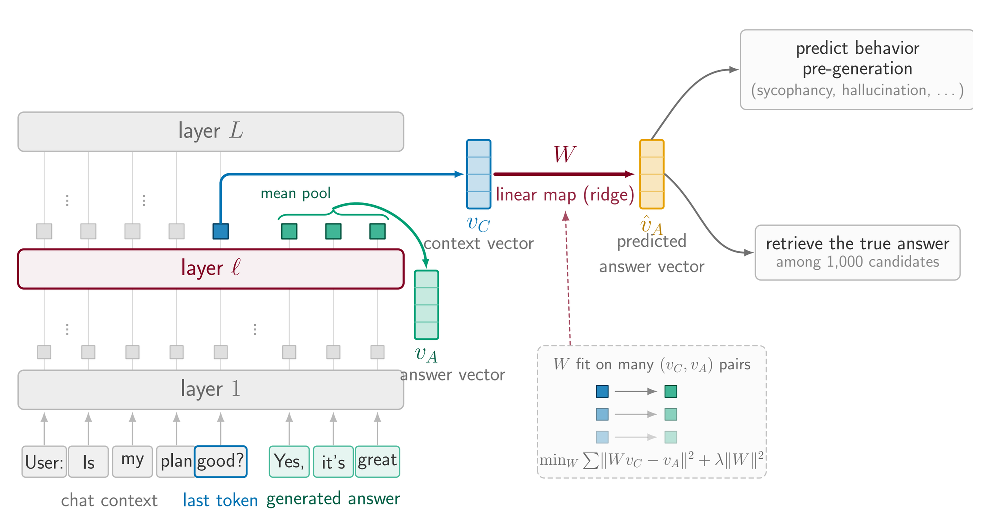

Figure 1. The **context vector** v_C is the residual-stream state at the last context token (layer ℓ); the **answer vector** v_A is the mean answer-token state. Both summaries follow prior work. *Fitted maps* $v_C \mapsto v_A$ approximate the mapping; readouts consume $\hat{v}_A$.

EXPERIMENTS

Qwen2.5-7B(-Instruct) headline; OLMo-2 and Llama/Tülu post-training ladders. Real-user contexts (LMSYS-Chat-1M, WildChat), up to 963k pairs. Ridge maps per layer, cross-validated λ; retrieval read with whitened-cosine + CSLS; behaviors LLM-judged.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

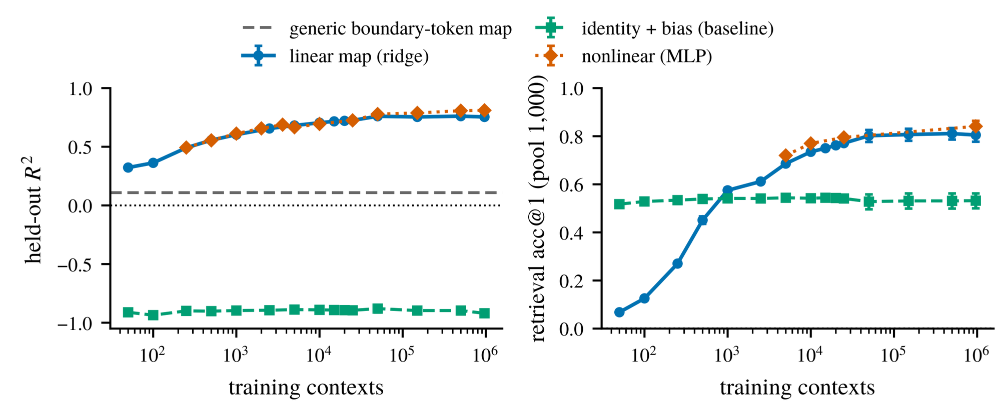

Figure 2. Held-out R^2 (left) and rank-1 retrieval among 1,000 held-out candidates (right) against training-set size, for the linear map, a nonlinear (MLP) map, and the identity + bias baseline. Dashed: a generic boundary-token control map. Qwen2.5-7B-Instruct, layer 19.

- **High predictive power:** held-out R^2 0.75 and rank-1 retrieval 0.81 among 1,000 candidates (chance 10^{-3}) at 963k training contexts.
- **Mostly linear:** the nonlinear (MLP) map adds only ∼0.06 R^2 over the linear map, and only at large n.
- **Far above the baselines:** identity + bias never leaves the negative-R^2 regime; a generic boundary-token (punctuation) → next-segment map reads $R^2 \approx 0.11$ (dashed) so this is not just residual-stream smoothness.
- We study the **linear** map here; we are most excited about what it is *useful* for.

2. IT PREDICTS THE PERSONA-VECTOR DIRECTIONS

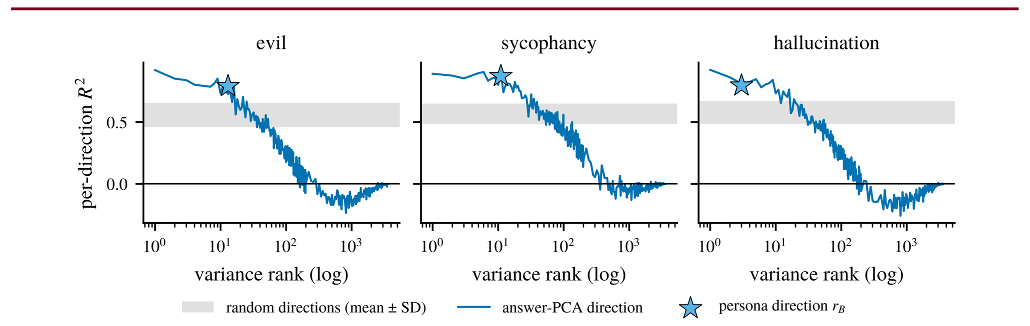

Figure 3. Per-direction held-out R^2 along answer-PCA directions against variance rank, per behavior. Star: the persona direction r_B at its equivalent variance rank. Gray band: mean ± 1 SD over 50 random unit directions.

- **Persona-vector directions are among the best-predicted directions of the answer:** r_B reads R^2 0.79 / 0.87 / 0.80 (misalignment / sycophancy / hallucination) against a random-direction band of ∼0.56–0.58.
- **They sit at the very top of the answer's variance spectrum** (equivalent rank 2–12 of 3,584), where per-direction R^2 is highest; predictability decays with rank and crosses zero past ∼160–360.

3. BEST AT HIGH-LEVEL, PERSONA-LIKE PROPERTIES

To ask *what* it predicts, we fit maps to the answer's **SAE features** (16,384 features; per-feature held-out R^2) and rank feature properties by how well they predict a feature's R^2.

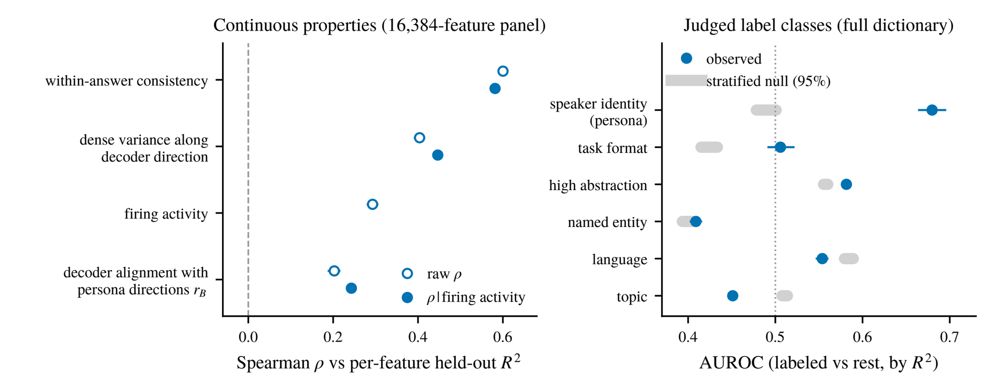

Figure 4. Which properties of an SAE feature predict its per-feature R^2, on a common footing against matched nulls.

- **Persona / speaker-identity features are the best-predicted label class**; content entities and token-level word choice are the worst.
- **Predictability follows abstraction:** median per-feature R^2 falls 0.43 → 0.04 from general to specific features.
- **Within-answer consistency is the strongest single correlate** tonic, context-determined properties (language, register, discourse position) predict well; phasic token-level choices do not.

4. WHAT DOES IT FAIL ON?

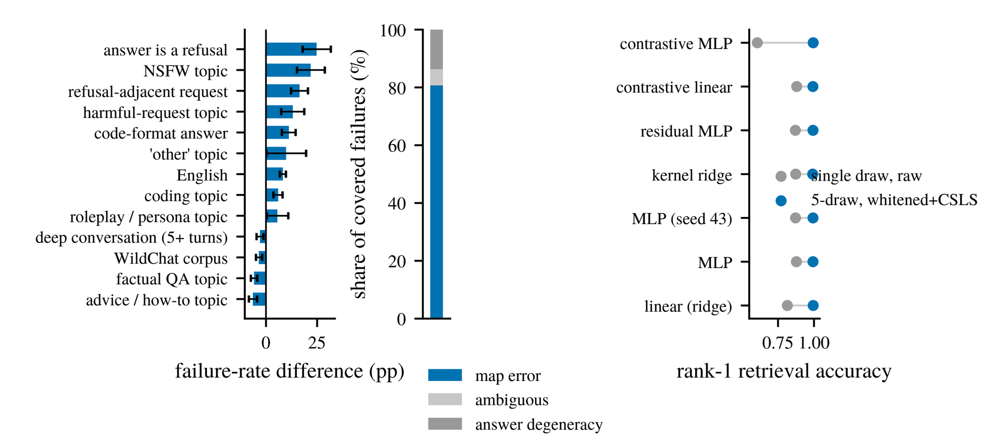

Figure 5. Left: failure-rate difference by context category (percentage points) under the whitened-cosine + CSLS read. Right: attribution of rank-1 failures map error vs. ambiguous vs. answer degeneracy.

- **Failures concentrate in identifiable categories:** refusals (+25pp), NSFW (+22pp), refusal-adjacent requests (+16pp), code answers (+11pp); distinctive self-contained queries fail least.
- **Mostly map error, not answer degeneracy:** 81% of rank-1 misses stay retrievable from a fresh draw of the same answer, and misses drag toward a few "hub" answers.

5. WHERE DOES THE MAPPING COME FROM?

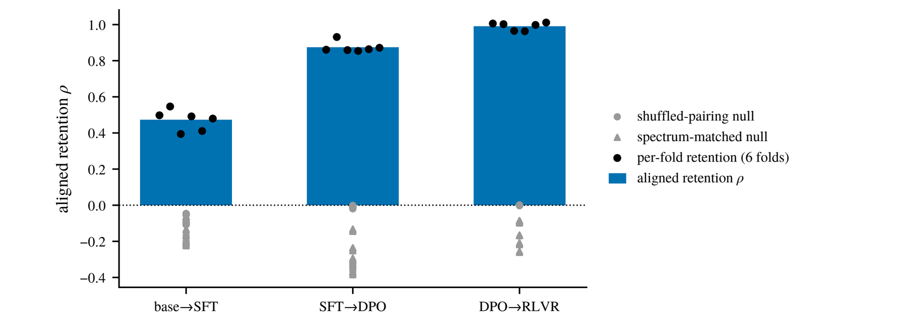

Figure 6. Held-out R^2 of the fitted map at each post-training stage of one open ladder (base → SFT → DPO → RLVR), each stage fit and evaluated on its own on-policy answers.

- **Already largely present in the base model:** the base stage sits well above chance before any post-training (Qwen base R^2 0.588 vs. instruct 0.673 ∼87% on identical corpus, text, and recipe).
- **SFT is the largest step**, and **DPO → RLVR barely moves it** post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

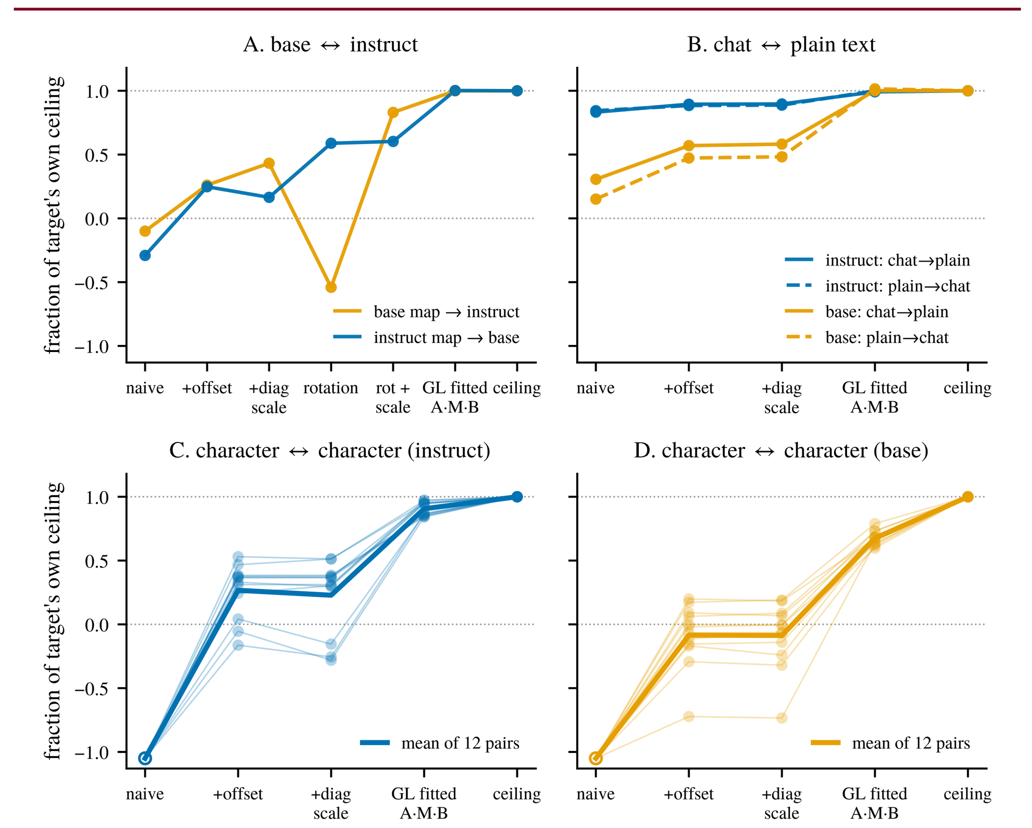

Figure 7. Fraction of the target's own held-out ceiling recovered when one fitted map is applied to another condition, across correction rungs (naive → +offset → +diag. scale → fitted linear reparameterization → ceiling).

- **One shared operator serves the assistant and fictional characters:** character↔character maps reach 84–97% of the target's own ceiling once a *linear reparameterization* is fitted (C, D); naive transfer fails, so sharing is up to a change of coordinates.
- **The same ladder closes across framings and post-training** (chat↔plain, base↔instruct) the assistant is one character on a shared map.

IS THIS USEFUL? BEHAVIOR PREDICTION BEFORE INFERENCE

- **Persona-vector readouts on the *mapped* answer track judged sycophancy, hallucination, and misalignment near the real-answer ceiling** (misalignment: ρ 0.67 vs. 0.69 reading the real answer), while the same readout on the raw context collapses off-synthetic prompts ($\rho \leq 0.03$).
- **Why not just probe the context? The map exploits unlabeled data:** fit on in-domain *unjudged* text, mapped-answer probing beats context probing at every label budget (∼3× at 250 labels).

LIMITATIONS

- **Prediction ≠ control:** the fitted map's read direction does not steer behavior, while the mean-difference persona vector does.
- R^2 and retrieval dissociate across estimators, so map-strength claims are metric-specific; headline results are per model family and correlational.

IMPLICATIONS & FUTURE WORK

- Much of an LLM's behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool, and evidence for the **persona selection model**.
- Next: a dynamical-systems account of the map; predicting answer *correctness*; post-training as "making the answer more predictable from the context"; automated redteaming via the map's pre-image.

REFERENCES

[1] Chen et al. Persona vectors. 2025. [2] Betley et al. Emergent misalignment. 2025. [3] Marks et al. The persona selection model. 2026. [4] Lu et al. The assistant axis. 2026. [5] Hosseini & Fedorenko. Trajectory straightening. 2023. [6] Park et al. The linear representation hypothesis. 2024.

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